

MODULE 08: INDUCTION & INDUCTANCE

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LECTURE 19

OUTLINE:

- Energy stored in a magnetic field
- Mutual induction
- LR circuit

ENERGY STORED IN A MAGNETIC FIELD

The energy stored in a magnetic field is a result of the magnetic field produced by a current-carrying conductor or an inductor. The amount of energy stored in the magnetic field depends on factors such as the **inductance** of the coil or conductor and the **amount of current** passing through it.

- The energy stored in the magnetic field is directly proportional to the square of the current passing through the conductor or coil.
- A higher inductance results in more energy storage for the same current.
- This energy is released when the magnetic field collapses or changes, such as, in the operation of electromagnetic devices like transformers, inductors.
- The energy stored in the magnetic field is released when the magnetic field changes due to changes in the current through the inductor.
 - When the current increases, energy is added to the magnetic field, and when the current decreases, energy is released from the field.
- This energy can also be transferred into other forms of energy, such as electrical energy, when a changing magnetic field induces an electromotive force (EMF) in a nearby conductor, according to Faraday's law of electromagnetic induction.

The energy stored by an inductor in a magnetic field can be calculated using the formula,

$$U_B = \left(\frac{1}{2}\right) i^2 L$$

where U_B represents the energy stored in the magnetic field, L is the inductance of the circuit, and i is the current flowing through the circuit.

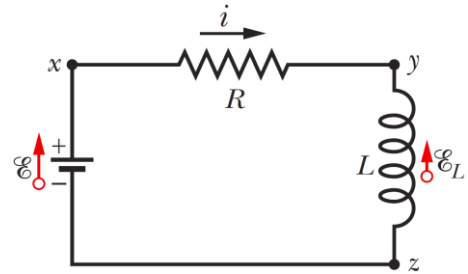
- The inductance (L) represents the ability of the inductor/circuit to store magnetic energy. It is determined by factors like the number of turns in the coil, the shape and size of the coil, and the relative permeability of the core material. The unit of inductance is the henry (H).
- The energy stored in the magnetic field is transient and can be released back into the circuit when the magnetic field collapses or changes. This phenomenon is utilized in devices such as inductors and transformers, where energy is stored in the magnetic field and then released or transferred to other parts of the circuit as needed.

Let us consider a source of emf ($\mathcal{E}$) connected to a resistor R and an inductor L . From the Kirchhoff's voltage law, we get,

$$\mathcal{E} - V_L - V_R = 0 \quad \Rightarrow \quad \mathcal{E} = L \frac{di}{dt} + iR$$

Multiplying each side of equation by i , we get,

$$i\mathcal{E} = iL \frac{di}{dt} + i^2 R$$



- The left side of the equation represents the rate at which the emf device delivers energy to the rest of the circuit [If a differential amount of charge dq passes through the battery of emf ($\mathcal{E}$) in time dt , the battery does work on it in the amount $\mathcal{E} dq$. The rate at which the battery does work is $\frac{\mathcal{E} dq}{dt}$, or $i\mathcal{E}$].
- The term $i^2 R$, in the equation represents the rate at which energy appears as thermal energy in the resistor.
- The remaining energy $iL \frac{di}{dt}$, that is delivered to the circuit but does not appear as thermal energy must, by the conservation-of-energy hypothesis, be stored in the magnetic field of the inductor.
 - Since, this equation represents the principle of conservation of energy for the RL circuits, the term must represent the rate $\frac{dU_B}{dt}$ at which magnetic potential energy U_B is stored in the magnetic field.

Therefore, we get,

$$\begin{aligned} \frac{dU_B}{dt} &= iL \frac{di}{dt} \\ \Rightarrow dU_B &= iL di \end{aligned}$$

Integrating yields,

$$\begin{aligned} \int_0^{U_B} dU_B &= \int_0^i iL di \\ \text{Or, } U_B &= \frac{1}{2} i^2 L \end{aligned}$$

which represents the total energy stored by an inductor L carrying a current i .

PROBLEM: A coil has an inductance of 53 mH and a resistance of 0.35Ω .

(a) If a 12 V emf is applied across the coil, how much energy is stored in the magnetic field after the current has built up to its equilibrium value?

(b) After how many time constants will half this equilibrium energy be stored in the magnetic field?

ANSWER: (a) The equilibrium current is given by,

$$i_{\max} = \frac{\mathcal{E}}{R} = \frac{12 \text{ V}}{0.35 \Omega} = 34.3 \text{ A}$$

Then the energy stored at equilibrium is given by,

$$U_{B\infty} = \frac{1}{2} i_{\max}^2 L = \left(\frac{1}{2}\right) (34.3 \text{ A})^2 (53 \times 10^{-3} \text{ H}) = 31 \text{ J}$$

(b) Given that, $U_B = \frac{1}{2} U_{B,max}$

$$\text{Therefore, we have, } \frac{1}{2} i^2 L = \frac{1}{2} \left(\frac{1}{2} i_{max}^2 L \right) \Rightarrow i = \left(\frac{1}{\sqrt{2}} \right) i_{max}$$

Thus, as the current increases from its initial value of 0 to its final value of i_{max} , the magnetic field will have half its final stored energy when the current has increased to $i = \left(\frac{1}{\sqrt{2}} \right) i_{max}$.

$$\text{We know, } i = \frac{\varepsilon}{R} (1 - e^{-t/\tau_L}) \text{ and } i_{max} = \frac{\varepsilon}{R}$$

So, the equation becomes,

$$\frac{\varepsilon}{R} \left(1 - e^{-\frac{t}{\tau_L}} \right) = \left(\frac{1}{\sqrt{2}} \right) i_{max} = \frac{\varepsilon}{\sqrt{2} R}$$

By canceling ε/R and rearranging, we can write this as

$$e^{-\frac{t}{\tau_L}} = 1 - \frac{1}{\sqrt{2}} = 0.293$$

which yields,

$$\frac{t}{\tau_L} = -\ln 0.293 = 1.23$$

$$\text{Or, } t \approx 1.2 \tau_L$$

Thus, the energy stored in the magnetic field of the coil by the current will reach half its equilibrium value 1.2 *time* constants after the emf is applied.

ENERGY DENSITY OF A MAGNETIC FIELD

Energy Density of a magnetic field is defined as the total amount of magnetic energy that is stored per unit volume within a region of space where a magnetic field exists.

- The energy density of a magnetic field quantifies the amount of energy stored in a given volume of space due to the presence of a magnetic field.
- It is denoted by u_B .
- SI unit is joules per cubic meter, J/m^3 .

Let us consider a length l near the middle of a long solenoid of cross-sectional area A carrying current i . The volume associated with this length is $V = Al$.

- The magnetic energy U_B stored by the length l of the solenoid must lie entirely within the volume of the solenoid, because the magnetic field outside such a solenoid is approximately zero.
- Moreover, the stored magnetic energy must be uniformly distributed within the solenoid as the magnetic field is uniform everywhere inside.

Thus, the energy stored per unit volume of the field is,

$$u_B = \frac{U_B}{V} = \frac{U_B}{Al} = \frac{i^2 L}{2Al} = \frac{L}{l} \frac{i^2}{2A} \quad \left[\because U_B = \frac{1}{2} i^2 L \right]$$

Here L is the inductance of length l of the solenoid.

Substituting $\frac{L}{l} = \mu_0 n^2 A$, where n is the number of turns per unit length of the solenoid, A being the cross-sectional area of a long solenoid, we get,

$$u_B = \frac{1}{2} \mu_0 n^2 i^2$$

We know that the magnetic field, $B = \mu_0 i n$. Therefore, the energy density of stored energy at any point where the magnitude of the magnetic field is B can be written as,

$$u_B = \frac{B^2}{2\mu_0}$$

CHECKPOINT: Table lists number of turns per unit length, current, and cross-sectional area for three solenoids. Rank solenoids according to magnetic energy density within them, greatest first.

Solenoid	Turns per Unit Length	Current	Area
<i>a</i>	$2n_1$	i_1	$2A_1$
<i>b</i>	n_1	$2i_1$	A_1
<i>c</i>	n_1	i_1	$6A_1$

ANSWER: *a and b tie, then c*

$$[u_B = \frac{1}{2} \mu_0 n^2 i^2]$$

MUTUAL INDUCTION

Mutual induction refers to the phenomenon where a changing magnetic field due to changing current in one coil induces an electromotive force (EMF) or voltage in an adjacent coil of wire. It is the basis for the operation of transformers and various types of electrical devices.

- Mutual induction occurs when two or more coils of wire are placed close to each other, and a changing current in one coil produces a varying magnetic field.
 - This changing magnetic field links with the adjacent coil, inducing a voltage or EMF in that coil.
 - The induced voltage is directly proportional to the rate of change of the magnetic field and the number of turns in the coil.
 - The induced voltage can be influenced by factors such as the magnetic permeability of the core material (if present), and the geometry and relative positions of the coils.

Figure a shows two circular close-packed coils near each other and sharing a common central axis. With the variable resistor set at a particular resistance R , the battery produces a steady current i_1 in *coil 1*. This current creates a magnetic field represented by the lines of $\vec{B}$ in the figure. *Coil 2* is connected to a galvanometer / ammeter. A magnetic flux Φ_{21} (the flux through *coil 2* associated with the current in *coil 1*) links the N_2 turns of *coil 2*.

The induced emf in *coil 2* is directly proportional to the current that passes through *coil 1*. Therefore,

$$N_2 \Phi_{21} \propto i_1 \quad \Rightarrow N_2 \Phi_{21} = M_{21} i_1 \dots \dots (1)$$

$$\Rightarrow M_{21} = \frac{N_2 \Phi_{21}}{i_1} \dots \dots (2)$$

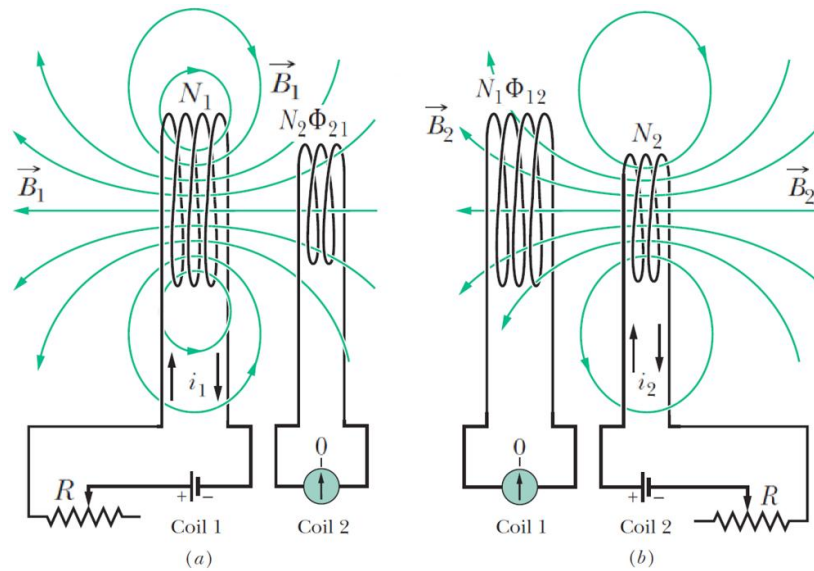


Fig. Mutual induction. (a) The magnetic field produced by current i_1 in coil 1 extends through coil 2. If i_1 is varied (by varying resistance R), an emf is induced in coil 2 and current registers on the meter connected to coil 2. (b) The roles of the coils interchanged.

If i_1 of the coil 1 is varied over time by varying R , by differentiating equation (1) we get,

$$M_{21} \frac{di_1}{dt} = N_2 \frac{d\Phi_{21}}{dt}$$

From Faraday's law, the magnitude of the emf ε_2 appearing in coil 2 due to the changing current in coil 1, can be given as (with a minus sign to indicate direction),

$$\varepsilon_2 = -N_2 \frac{d\Phi_{21}}{dt} = -M_{21} \frac{di_1}{dt} \quad \left[\because \varepsilon_2 = -N_2 \frac{d\Phi_{21}}{dt} \right]$$

Similarly, by interchanging the roles of coils 1 and coil 2, as in Fig. b; a current i_2 in coil 2 by means of a battery is set up, which produces a magnetic flux Φ_{12} that links coil 1. If i_2 is varied with time by varying R , we get,

$$\varepsilon_1 = -M_{12} \frac{di_2}{dt}$$

Thus, the emf induced in either coil is proportional to the rate of change of current in the other coil. Here, M_{21} and M_{12} are the proportionality constants. Since, they are equal, we get,

$$M_{21} = M_{12} = M$$

Therefore,

$$\varepsilon_1 = -M \frac{di_2}{dt}$$

$$\varepsilon_2 = -M \frac{di_1}{dt}$$

Mutual induction is the fundamental principle behind the operation of transformers.

- Transformers consist of two or more coils of wire wound around a common magnetic core.
- When an alternating current (AC) flows through the primary coil, it produces a changing magnetic field.
- This changing magnetic field induces a voltage in the secondary coil, allowing the transfer of electrical energy from one coil to another.

PROBLEM: Figure shows two circular close-packed coils, the smaller (radius R_2 , with N_2 turns) being coaxial with the larger (radius R_1 , with N_1 turns) and in the same plane.

(a) Derive an expression for the mutual inductance M for this arrangement of these two coils, assuming that $R_1 \gg R_2$.

(b) What is the value of M for $N_1 = N_2 = 1200$ turns, $R_2 = 1.1$ cm, and $R_1 = 15$ cm?

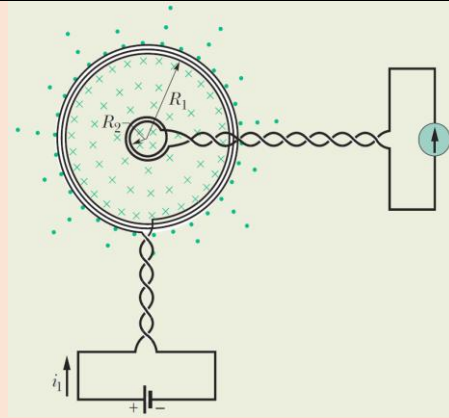


Fig. A small coil is located at the center of a large coil. The mutual inductance of the coils can be determined by sending current i_1 through the large coil.

ANSWER: (a) The magnetic field through the larger coil due to the smaller coil is nonuniform in both magnitude and direction; So, the flux through the larger coil due to the smaller coil is nonuniform and difficult to calculate. However, the smaller coil is small enough for us to assume that the magnetic field through it due to the larger coil is approximately uniform. Thus, the flux through it due to the larger coil is also approximately uniform. Hence, to find M we shall assume a current i_1 in the larger coil and calculate the flux linkage $N_2 \Phi_{21}$ in the smaller coil,

$$M = \frac{N_2 \Phi_{21}}{i_1}$$

The flux Φ_{21} through each turn of the smaller coil is,

$$\Phi_{21} = B_1 A_2$$

where B_1 is the magnitude of the magnetic field at points within the small coil due to the larger coil and is the area enclosed by the turn. Thus, the flux linkage in the smaller coil (with its N_2 turns) is

$$N_2 \Phi_{21} = N_2 B_1 A_2$$

The B_1 at points within the smaller coil, can be obtain using,

$$B(z) = \frac{\mu_0 i R^2}{2(R^2 + z^2)^{3/2}}$$

with z set to 0 because the smaller coil is in the plane of the larger coil. That equation tells us that each turn of the larger coil produces a magnetic field of magnitude $\frac{\mu_0 i_1}{2R_1}$ at points within the smaller coil. Thus, the larger coil (with its N_1 turns) produces a total magnetic field of magnitude,

$$B_1 = N_1 \frac{\mu_0 i_1}{2R_1}$$

at points within the smaller coil. Substituting equation for B_1 and πR_2^2 for A_2 yields,

$$N_2 \Phi_{21} = N_2 \left(N_1 \frac{\mu_0 i_1}{2R_1} \right) (\pi R_2^2) = \frac{\pi \mu_0 N_1 N_2 R_2^2}{2R_1} i_1 \quad \Rightarrow M = \frac{N_2 \Phi_{21}}{i_1} = \frac{\pi \mu_0 N_1 N_2 R_2^2}{2R_1}$$

(b) We know,

$$M = \frac{\pi \mu_0 N_1 N_2 R_2^2}{2R_1} = \frac{\pi \left(4\pi \times 10^{-7} \frac{H}{m} \right) (1200)(1200)(0.011 \text{ m})^2}{2(0.15 \text{ m})} = 2.29 \times 10^{-3} \text{ H}$$

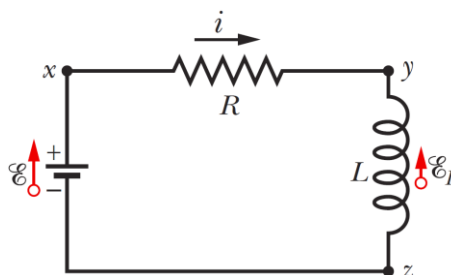
$$M \approx 2.3 \text{ mH}$$

LR CIRCUIT

An LR circuit, also known as an inductor-resistor circuit, is an electrical circuit that consists of a resistor (R) and an inductor (L) connected in series. The inductor is a passive electronic component that stores electrical energy in its magnetic field when current flows through it, while the resistor opposes the flow of current in the circuit.

When a voltage source is connected to an LR circuit

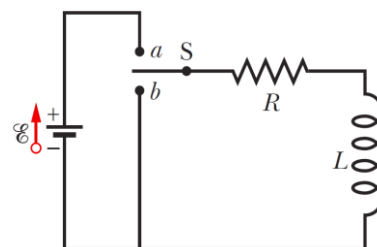
- The *behavior of the LR circuit* depends on the values of the resistor, inductor, and applied voltage.
- When voltage is applied to the circuit initially, the current does not change instantaneously due to the inductor's property of opposing changes in current. Due to back emf formation.
- The rate at which the current changes is determined by the inductance of the inductor (L) and the resistance in the circuit (R). Time taken for current to reach its maximum or minimum value is known as time constant (τ) of the RL circuit and is given by, $\tau = \frac{L}{R}$.
- As time progresses, the inductor's magnetic field builds up, and the current approaches a steady-state value determined by the resistance and the applied voltage.
- Voltage across the inductor is proportional to the rate of change of current, and expressed by $V_L = L \left(\frac{di}{dt} \right)$, where V is the voltage, L is the inductance, and $\frac{di}{dt}$ is the rate of change of current.



Charging of Inductor:

Let us consider a single-loop RL circuit with source emf (ϵ) consists of a resistor R and an inductor L . When the switch S in Fig. is closed on a , for example,

- The current in the resistor starts to rise.
 - If the inductor were not present, the current would rise rapidly to a steady value ϵ/R .
- A self-induced emf (ϵ_L) appears in the circuit because of the inductor. The change of i leads to induced emf which opposes the rise of the current. Thus, the induced emf opposes the battery emf ϵ in polarity.
- Therefore, the current through the resistor depends on the difference between two emfs ($\epsilon - \epsilon_L$), namely, a constant emf (ϵ) due to the battery and a variable $\epsilon_L \left(= -L \frac{di}{dt} \right)$ due to self-induction.
- As long as this ϵ_L is present, the current remains less than $\frac{\epsilon}{R}$.



With time the rate at which the current increases become less rapid and the magnitude of the self-induced emf, which is proportional to $\frac{di}{dt}$, becomes smaller. Thus, the current in the circuit approaches ϵ/R asymptotically. Thus, initially, an inductor acts to oppose changes in the current through it. After a long time, it acts like ordinary connecting wire.

If a constant emf (ε) is applied to a circuit containing R and L , the current does not rise instantly. Instead, it increases exponentially toward a maximum value of $i_0 = i_{max} = \frac{\varepsilon}{R}$. As the inductor opposing the change in current, during this rise, the voltage across the inductor decays exponentially from V to 0. From the Kirchhoff's voltage law, we get,

$$\varepsilon = \varepsilon_R + \varepsilon_L = iR + L \frac{di}{dt} \Rightarrow \frac{di}{dt} + \frac{R}{L}i = \frac{\varepsilon}{L}$$

This differential equation involving variable i and its first derivative di/dt . The solution gives us,

$$i(t) = \frac{\varepsilon}{R} \left(1 - e^{-\frac{Rt}{L}}\right) = \frac{\varepsilon}{R} \left(1 - e^{-\frac{t}{\tau_L}}\right)$$

$$i(t) = i_0(1 - e^{-t/\tau_L})$$

Here τ_L is the inductive time constant ($\tau_L = \frac{L}{R}$).

Therefore,

$$\varepsilon_L(t) = L \frac{di}{dt} = L \frac{d}{dt} \left[\frac{\varepsilon}{R} \left(1 - e^{-\frac{Rt}{L}}\right) \right] = L \left(\frac{\varepsilon}{R} \right) \left(\frac{R}{L} \right) e^{-Rt/L} = \varepsilon e^{-t/\tau_L}$$

Discharging of Inductor:

when the constant emf ε_0 is removed (or disconnected) by switching S in Figure thrown to terminal b of the circuit, the current through the resistor begins to decrease/decay exponentially from i_0 toward zero. However, it cannot drop immediately to zero but must decay to zero over time. At that moment using Kirchhoff's law we get (by putting $\varepsilon = 0$),

$$\varepsilon_R + \varepsilon_L = 0 \Rightarrow iR + L \frac{di}{dt} = 0 \Rightarrow \frac{di}{dt} + \frac{R}{L}i = 0$$

During discharge, the established initial current is $i_0 = \frac{\varepsilon_0}{R}$. Thus, by solving this differential equation we get,

$$i = i_0 e^{-Rt/L} = i_0 e^{-t/\tau_L} = \frac{\varepsilon_0}{R} e^{-t/\tau_L}$$

Therefore, both the current rise and current decay in an RL circuit are governed by the inductive time constant, τ_L .

The inductor, in response, generates a back EMF to sustain the current, causing the voltage across it to decay from $-V_0$ to 0, given by

$$\varepsilon_L(t) = L \frac{di}{dt} = L \frac{d}{dt} \left[\frac{\varepsilon_0}{R} e^{-\frac{Rt}{L}} \right] = -L \left(\frac{\varepsilon_0}{R} \right) \left(\frac{R}{L} \right) e^{-t/\tau_L} = -\varepsilon_0 e^{-t/\tau_L}$$

Impedance of LR Circuit:

The impedance (Z) of an RL circuit is a measure of its opposition to the flow of alternating current. The impedance of the RL series circuit can be written as, $Z = R + jX_L$.

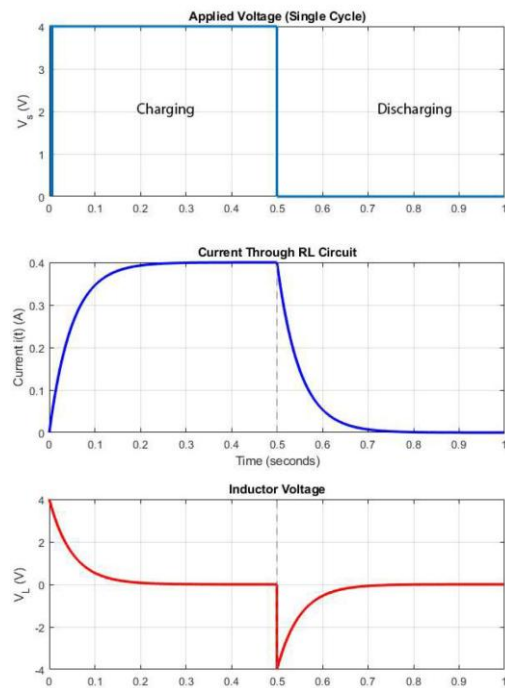


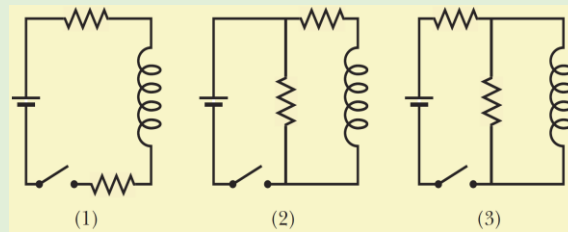
Fig 1: I-V waveform with respect to time

The impedance of series RL circuit is nothing but the combine effect of resistance (R) and inductive reactance (X_L). The magnitude of impedance Z in ohms is given by, $|Z| = \sqrt{R^2 + X_L^2}$ and from right angle triangle, phase angle,

$$\theta = \tan^{-1} \left(\frac{X_L}{R} \right)$$

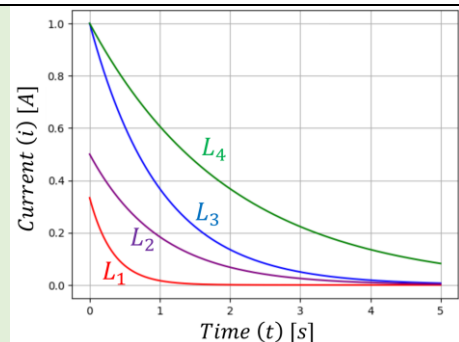
Here R is the resistance, and X_L refers to the inductive reactance. We know, $X_L = \omega L$. Therefore, $|Z| = \sqrt{R^2 + \omega^2 L^2}$.

CHECKPOINT: The figure shows three circuits with identical batteries, inductors, and resistors. Rank the circuits according to the current through the battery (a) just after the switch is closed and (b) a long time later, greatest first.



ANSWER: (a) 2, 3, 1 (zero); (b) 2, 3, 1

CHECKPOINT: A series RL circuit has been connected to a switch that remained closed (switch ON mode) for a long time and is then opened (switch OFF mode). The resulting current through the inductor is shown in the figure for four different sets of values for the resistance R and inductance L : (i) R_0 and L_0 , (ii) $3R_0$ and L_0 , (iii) R_0 and $2L_0$, (iv) $2R_0$ and $2L_0$. Which set goes with which curve?



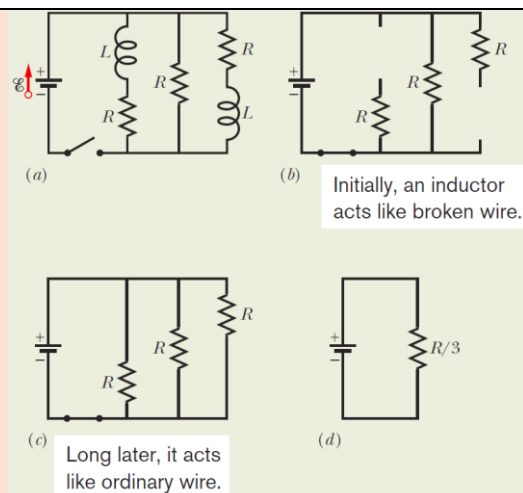
ANSWER: (i) L_3 ; (ii) L_1 ; (iii) L_4 ; (iv) L_2 ;

PROBLEM: Figure *a* shows a circuit that contains three identical resistors with resistance $R = 9.0 \, \Omega$, two identical inductors with inductance $L = 2.0 \, \text{mH}$, and an ideal battery with emf $\mathcal{E} = 18 \, \text{V}$.

(a) What is the current i through the battery just after the switch is closed?

(b) What is the current i through the battery long after the switch has been closed?

ANSWER: (a) The current through each inductor is zero before the switch is closed, and it is also zero just afterward.



Thus, immediately after the switch is closed, the inductors act as broken wires, as indicated in Fig. *b*. We then have a single-loop circuit for which the loop rule gives us,

$$\varepsilon - iR = 0 \quad \Rightarrow i = \frac{\varepsilon}{R} = \frac{18 \text{ V}}{9.0 \Omega} = 2.0 \text{ A}$$

(b) The circuit consists of three identical resistors in parallel. From equation, their equivalent resistance is $R_{eq} = \frac{R}{3} = \frac{9.0 \Omega}{3} = 3.0 \Omega$. From equivalent circuit diagram (Fig. d),

$$\varepsilon - iR_{eq} = 0 \quad \Rightarrow i = \frac{\varepsilon}{R_{eq}} = \frac{18 \text{ V}}{3.0 \Omega} = 6.0 \text{ A}$$

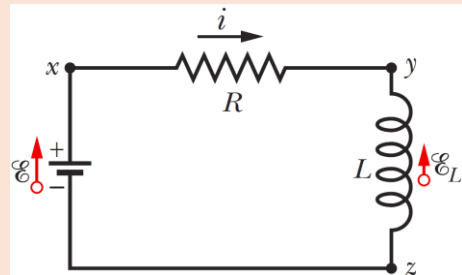
PROBLEM: A solenoid has an inductance of 53 mH and a resistance of 0.37Ω . If the solenoid is connected to a battery, how long will the current take to reach half its final equilibrium value? (This is a *real solenoid* because we are considering its small, but nonzero, internal resistance.)

ANSWER: From the circuit we get,

$$L \frac{di}{dt} + iR = \varepsilon$$

This is a differential equation which gives,

$$i = \frac{\varepsilon}{R} e^{-t/\tau_L} = i_0 e^{-t/\tau_L}$$



According to that solution, current i increases exponentially from zero to its final equilibrium value of ε/R . Let t_0 be the time that current i takes to reach half its equilibrium value.

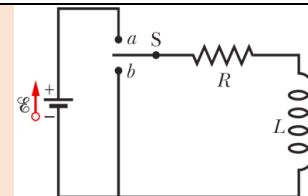
At half of its final equilibrium value, the current $i = \frac{1}{2} \frac{\varepsilon}{R}$

$$\Rightarrow \frac{1}{2} \frac{\varepsilon}{R} = \frac{\varepsilon}{R} (1 - e^{-t_0/\tau_L})$$

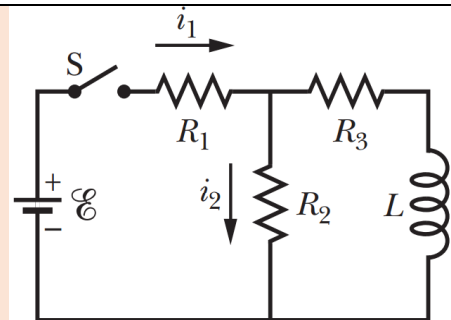
By canceling ε/R , isolating the exponential, and taking natural logarithm of each side. We get,

$$t_0 = \tau_L \ln 2 = \frac{L}{R} \ln 2 = \frac{53 \times 10^{-3} \text{ H}}{0.37 \Omega} \ln 2 = 0.10 \text{ s}$$

PROBLEM 30-52: The switch in figure is closed on a at time $t = 0$. What is the ratio $\varepsilon_L/\varepsilon$ of the inductor's self-induced emf to the battery's emf (a) just after $t = 0$ and (b) at $t = 2.00\tau_L$? (c) At what multiple of τ_L will $\frac{\varepsilon_L}{\varepsilon} = 0.500$?



PROBLEM 30-54: In Fig., $\varepsilon = 100 \text{ V}$, $R_1 = 10.0 \Omega$, $R_2 = 20.0 \Omega$, $R_3 = 30.0 \Omega$, and $L = 2.00 \text{ H}$. Immediately after switch S is closed, what are (a) i_1 and (b) i_2 ? (Let currents in the indicated directions have positive values and currents in the opposite directions have negative values.) A long time later, what are (c) i_1 and (d) i_2 ? The switch is then reopened. Just then, what are (e) i_1 and (f) i_2 ? A long time later, what are (g) i_1 and (h) i_2 ?



PROBLEM 30-58: Suppose the emf of the battery in the circuit shown in figure varies with time t so that the current is given by $i(t) = 3.0 + 5.0t$, where i is in amperes and t is in seconds. Take $R = 4.0\ \Omega$ and $L = 6.0\ H$, and find an expression for the battery emf as a function of t .

